Math 190-1: Mathematics of Sustainability	**Spring 2026**

Lecture 6: Mixed Strategies

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Lecturer: Mitchell Scott

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6.1 Mixed-strategies

Recall the game:

		Player 2	
		L	R
Player 1	U	3,0	0,0
	M	0,0	3, 0
	D	1, 0	1, 0

We used this to show that the set of IESDS-equilbria (which was all stratgy profiles) was a bigger collection that the set of rationalizable strategies (D isn't rationalizable). However, what if a player flips a fair coin to decide what strategy to do. If H, then play U, and if T, play M. We calculate the following expected payoff.

$$\mathbb{E}(u(L)) = \mathbb{P}(H)u(L|U) + \mathbb{P}(T)u(L|M) = \frac{1}{2}3 + \frac{1}{2}0 = \frac{3}{2}$$

$\mathbb{E}(u(M)) = \frac{1}{2}0 + \frac{1}{2}3 = \frac{3}{2}$. So L has an expected payoff of $\frac{3}{2}, 0$ and R has a payoff of $\frac{3}{2}, 0$. This dominates D, which means that we can get rid of it!.

This means that sometimes it would be better to randomize your strategies.

Example 6.1 *Imagine playing rock, paper, scissors with someone who only throws R. How boring of a game. You would always choose P to cover their rock. Ideally your opponent would roll a 3 sided die and play a randomized strategy for rock paper scissors.*

Let's call this strategy $\sigma = (\sigma(R), \sigma(P), \sigma(S)) = (\frac{1}{2}, \frac{1}{2}, 0)$. *As long as* $\sigma(R), \sigma(P), \sigma(S) \geq 0, \sigma(R) + \sigma(P) + \sigma(S) = 1$, *it is a valid mixed-strategy profile. But how do figure out the expected payoff of this kind.*

Let Player 2 play $\sigma_2 = (\frac{1}{2}, \frac{1}{2}, 0)$. *What should player 1 do?*

$$\begin{aligned}
v_1(R, \sigma_2) &= \frac{1}{2}(0) + \frac{1}{2}(-1) + 0(1) = -\frac{1}{2} \\
v_1(S, \sigma_2) &= \frac{1}{2}(1) + \frac{1}{2}(0) + 0(-1) = \frac{1}{2} \\
v_1(P, \sigma_2) &= \frac{1}{2}(-1) + \frac{1}{2}(1) + 0(1) = 0
\end{aligned}$$

This means that $\mathrm{BR}_1(\sigma_2) = \{P\}$*, which shouldn't surprise you. If you knew your opponent never played Scissors* S*, then Paper* P *can never be cut, at worse it ties (two papers), and at best it wins.*

Example 6.2 *Let's play a game. Both you and your friend have pennies, and you get to decide which side you want up. If you both play heads or both play tails, player 1 gets both pennies. If the two pennies don't match, then player 2 gets both pennies. What should we do?*

		Player 2 H	T
Player 1	H	1, -1	-1, 1
	T	-1, 1	1, -1

Table 6.1: Matching Pennies.

There is clearly not a dominating strategy, and no pure strategy Nash equilbrium.

$$\mathrm{BR}_1(H) = \{H\}\,\mathrm{BR}_2(H) = \{T\}$$
$$\mathrm{BR}_1(T) = \{T\}\,\mathrm{BR}_2(T) = \{H\},$$

since no strategies are when both players are mutually choosing a best response. To find a Mixed-strategy Nash Equilbrium, we let $p = \sigma_1(H) \implies (1-p) = \sigma_1(T)$*. Similarly, we let* $q = \sigma_2(H) \implies (1-q) = \sigma_2(T)$*.*

Now to start we will find the $\mathrm{BR}_1(\sigma_2)$

$$\begin{aligned} v_1(H,\sigma_2) &= \sigma_2(H)v_1(H,H) + \sigma_2(T)v_1(H,T) \\ &= q \cdot 1 + (1-q)\cdot(-1) \\ &= 2q - 1 \end{aligned}$$

Now for the best response if Player 1 chooses T*.*

$$\begin{aligned} v_1(T,\sigma_2) &= \sigma_2(H)v_1(T,H) + \sigma_2(T)v_1(T,T) \\ &= q \cdot (-1) + (1-q)\cdot(1) \\ &= 1 - 2q \end{aligned}$$

This means we want to maximize the playoff, so if $v_1(H,\sigma_2) > v_1(T,\sigma_2)$*, then we will choose* H*. This happens when* $q > \frac{1}{2}$ *since* $v_1(H) > 0 > v_1(T)$*.*

We can also see that if $q < \frac{1}{2}$ *then* $v_1(T) > v_1(H)$*. When* $q = \frac{1}{2}, v_1(H) = v_1(T)$*.*

This means that

$$\mathrm{BR}_1(\sigma_2) = \begin{cases} T, & q < \frac{1}{2} \\ H, & q > \frac{1}{2} \\ 0 \le p \le 1, & q = \frac{1}{2} \end{cases}.$$

Now let's look at the best response for Player 2 responding to σ_1*.*

$$\begin{aligned} v_2(\sigma_1,H) &= \sigma_1(H)v_2(H,H) + \sigma_2(T)v_2(T,H) \\ &= p \cdot (-1) + (1-p)\cdot(1) \\ &= 1 - 2p \end{aligned}$$

and

$$\begin{aligned} v_2(\sigma_1, T) &= \sigma_1(H)v_2(H,T) + \sigma_2(T)v_2(T,T) \\ &= p \cdot (1) + (1-p) \cdot (-1) \\ &= 2p - 1 \end{aligned}$$

so their best response profile will be

$$\mathrm{BR}_2(\sigma_1) = \begin{cases} H, & p \leq \frac{1}{2} \\ T, & p > \frac{1}{2} \\ 0 \leq q \leq 1, & p = \frac{1}{2} \end{cases}$$

This means the Nash equilbrium is when $\mathrm{BR}_1(\sigma_2) = \mathrm{BR}_2(\sigma_1)$*, which happens when* $p = (1-p) = \frac{1}{2} = q = (1-q)$*, or we can say the set of Nash equilbrium is* $\{(\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2})\} = \{(\sigma_1^*, \sigma_2^*)\}$.

Remark 6.3 *We notice that* $v_1(H, \sigma_2^*) = v_1(T, \sigma_2^*)$ *and* $v_2(\sigma_1^*, H) = v_2(\sigma_1^*, T)$*. This is not a coincidence.*

Remark 6.4 *A pure strategy is a special case of a mixed strategy, where* $\sigma(s) = 1$ *for one* s*, a pure strategy.*

Remark 6.5 *A player who plays a mixed strategy in a Nash Equilbrium doesn't do that for the (direct) benefit, the payoff of all purestrategies that they are mixing between are all the same. Instead they do this so that their opponent doesn't take advantage of the deviation from the Nash equilblrium. In otherwords, Player 1 plays mixed strategy such that Player 2 is indifferent between the pure0strategies in support of the optimal mixed strategy* σ_2^*.

Example 6.6 (Combining Pure and Mixed Nash Equilbria) *Consider the game below:*

		Player 2	
		C	R
Player 1	M	0, 0	3, 5
	D	4, 4	0, 3

We can see that:

$$\begin{aligned} &\mathrm{BR}_1(C) = \{D\} \quad \mathrm{BR}_1(R) = \{M\} \\ &\mathrm{BR}_2(D) = \{C\} \quad \mathrm{BR}_2(M) = \{R\} \end{aligned}$$

This means that the pure strategy Nash equilibria are (C, D) *and* (R, M)*. But there is a theorem that says the total number of mixed-strategy Nash equilibria is odd, which means there has to be another mixed strategy Nash Equilbrium. Let's find it.*

Let's say that it is (σ_1^*, σ_2^*) *is a Nash equilbrium, where* σ_i^* *is the mixed Nash equilbrium for player* i*. This means that* $v_1(M, \sigma_2^*) = v_1(D, \sigma_2^*)$ *and* $v_2(\sigma_1^*, C) = v_2(\sigma_1^*, R)$*. To find this we need to say that in Player 2's Nash equilbrium, they will pick* C q *percent of the time, which means they will pick* R $(1-q)$ *percent of the time.*

$$\begin{aligned} v_1(M, \sigma_2^*) &= \sigma_2^*(C)v_1(M, C) + \sigma_2^*(R)v_1(M, R) \\ &= q \cdot (0) + (1-q) \cdot 3 \\ &= 3 - 3q \\ v_1(D, \sigma_2^*) &= \sigma_2^*(C)v_1(D, C) + \sigma_2^*(R)v_1(D, R) \\ &= q \cdot (4) + (1-q) \cdot 0 \\ &= 4q \end{aligned}$$

Since we know the Nash Equilbrum is happening when $v_1(M, \sigma_2^*) = v_1(D, \sigma_2^*)$*, we set these two equations equal to each other, to see*

$$\begin{aligned} 3 - 3q &= 4q \\ 3 &= 7q \\ \frac{3}{7} &= q. \end{aligned}$$

Now we turn our attention to what Player 2 should do with Player 1's strategy $\sigma_1^* = (\sigma_1^*(M), \sigma_1^*(D)) = (p, 1-p)$.

$$\begin{aligned} v_2(\sigma_1^*, C) &= 0 \cdot p + (1-p) \cdot 4 \\ &= 4 - 4p \\ v_2(\sigma_1^*, R) &= 5 \cdot p + (1-p) \cdot 3 \\ &= 2p + 3 \\ 4 - 4p &= 2p + 3 \\ 1 &= 6p \\ \frac{1}{6} &= p \end{aligned}$$

Putting this all together, we see that $(\sigma_1^*, \sigma_2^*) = \Big(((\frac{1}{6}, \frac{5}{6}), (\frac{3}{7}, \frac{4}{7}) \Big)$

We are now going to see how IESDS and Rationalizability work with mixed strategies.

Example 6.7 *Consider the example*

		Player 2		
		L	C	R
Player 1	U	5, 1	1, 4	1, 0
	M	3, 2	0, 0	3, 5
	D	4, 3	4, 4	0, 3

We are going to try to find an IESDS strategy. Consider Player 2's mixed strategy $\sigma_2 = \big(\sigma_2(L), \sigma_2(C), \sigma_2(R)\big) = \big(0, \frac{1}{2}, \frac{1}{2}\big)$

We can verify that this strictly dominates L

$$v_2(U, \sigma_2) = 0 \cdot 1 + \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 0 = 2.0 > 1 = v_2(U, L)$$
$$v_2(M, \sigma_2) = 0 \cdot 2 + \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 5 = 2.5 > 2 = v_2(M, L)$$
$$v_2(D, \sigma_2) = 0 \cdot 3 + \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 3 = 3.5 > 3 = v_2(D, L)$$

This means we eliminate L, as it is strictly dominated which means it is never a best response.

		Player 2	
		C	R
Player 1	U	1, 4	1, 0
	M	0, 0	3, 5
	D	4, 4	0, 3

We see that no pure strategy dominates another pure strategy from either player. Let's check if there is a mixed strategy that we can find. Consider if Player 1 mixes between M and D so $\sigma_1 = (\sigma_1(U), \sigma_1(M), \sigma_1(D)) = \left(0, \frac{1}{2}, \frac{1}{2}\right)$.

$$v_1(\sigma_1, C) = 0 \cdot 1 + \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 4 = 2.0 > 1 = v_1(U, C)$$
$$v_1(\sigma_1, R) = 0 \cdot 1 + \frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 0 = 1.5 > 1 = v_1(U, R)$$

This tells us to eliminate U since it is dominated by σ_1*. This leads us to*

		Player 2	
		C	R
Player 1	M	0, 0	3, 5
	D	4, 4	0, 3

This is all that can be eliminated. So the IESDS strategies are $\{M, D\}$ *for Player 1 and* $\{C, R\}$ *for Player 2.*

6.1.1 Summary

Remark 6.8 (IESDS Equilbrium) IESDS strategies *are those that remain after iteratively eliminating strictly dominated strategies.*

Remark 6.9 (Rationalizable Equilbrium) Rationalizable strategies *are those that remain after iteratively eliminating strategies that are never-a-best-response.*

Remark 6.10 (Nash Equilbrium) *At the risk of being repetitive, let me emphasize the requirements for a Nash equilibrium:*

1. *Each player is playing a best response to his beliefs.*
2. *The beliefs of the players about their opponents are correct*

Theorem 6.11 (Nash's Existence Theorem) *Any n-player normal-form game with finite strategy sets S_i for all players has a (Nash) equilibrium in mixed strategies.*